

Fact sheet on applications for GCS/HLRS computing time projects on Hunter

1. Prospective users can apply for computing time on Hunter. Depending on the demand of compute time, there are two different ways for the application:
 - Applications for Large-Scale projects requiring more than 25.000 Hunter-node-hours in a one year time frame can apply twice a year at defined periods of time. The dates for the next call can be found on the [GCS webpage](#). The call deadlines are strict deadlines, applications submitted after the call deadline will be rejected.
 - Applications for smaller projects can be submitted at any time. Please find instructions on the [HLRS web page](#).

Applications are accepted **online only**. Applications via email **will not be accepted**.

2. Scientists with a high demand for computing time **are explicitly encouraged to apply for a GCS-Large Scale project**. If the application is not granted as a GCS-LS project it is automatically considered as a regular HLRS project. Therefore, **there are no disadvantages for the applicants** submitting a large-scale project.
3. Eligible are scientists employed at universities or research facilities in Germany. Here, the nationality of the applicant ("Principal investigator, PI") does not play a role. The PI of a project must have a proven scientific record (preferable a PhD or comparable degree) and must be able to successfully accomplish the proposed tasks.
4. Available Systems

Hunter is the new HPC system at HLRS. Hunter is based on the AMD MI300A accelerator combining CPU and GPU cores in one socket. Applicants have to make sure to have their application(s) running on Hunter efficiently. Test access for application porting and efficiency tests is available. Support for application porting is also available. If not yet done so, please get in touch with the HLRS team for that.

Overview Hunter

Hunter (Accelerated system with Dragonfly network)	
Avail. Hunter-node-h/call	Approx.. 450.000
Processor	AMD® MI300A; each APU provides 24 Zen4 cores and 228 accelerator CUs
Nodes	188, four AMD300A APUs per node
Memory	HBM3; 512 GB/node; 21,2 TB/s memory bandwidth per node
Interconnect	Slingshot 11, 200 Gb/s per link, four links per node

5. Important Remarks
 - Hunter will provide a fast and expensive interconnect as it is a homogenous HPC system for scaling applications. Therefore
 - a good scalability of the application has to be shown
 - job farming is not an appropriate way to use a large number of nodes in parallel on that system.
 - Hunter is a step stone system for a much larger successor system to become available in 2027.
 - The file system is a shared resource. Not well behaving applications have a significant impact on other running applications. Therefore
 - please provide information about your I/O strategy
 - make sure not to use a large number of small files with your application

- Regular operation includes waiting times for jobs as well as maintenance phases. Therefore, please make sure to be able to execute the job set according to your work plan in about 2000 hours wall clock time per year.
6. For the preparation of the requested project description, please **use the template** which is available in [Word](#), [LaTeX](#) and [PDF](#) and adhere to the [guidelines](#) given on the GCS webpage. Since the templates may change from call to call make sure to use always the latest version using the links above. Please be aware, that the size of the description is limited to **18 pages (font size 11pt) and 60 MB**. In case you apply for a project extension, please also upload the status report (max. 10 pages) **as a separate file** (PDF). Please **use the template** which is available in [Word](#), [LaTeX](#) and [PDF](#).
 7. For more details please refer to the [HLRS web site](#)